Chemical Reactions and Equations

Detailed Study Notes

Chapter 1 • Class X Science • NCERT

■ Topics Covered	Chemical Equations · Types of Reactions · Oxidation & Reduction · Corrosion · Rancidity
■ Key Concepts	Balancing equations, Combination, Decomposition, Displacement, Double Displacement, Red
■■ Activities	11 hands-on activities supporting each concept

1. Introduction to Chemical Reactions

A **chemical reaction** is a process in which the nature and identity of substances change. Unlike physical changes, chemical changes result in the formation of entirely new substances with different properties.

Examples from daily life:

■	Milk turning sour at room temperature
■	Iron pan/nail rusting in humid air
■	Grapes undergoing fermentation
■	Food being cooked
■	Food digestion in the body
■■■	Respiration (breathing)

Indicators that a Chemical Reaction has occurred:

■ Change in Colour	e.g. brown iron nail in copper sulphate solution
■■ Change in Temperature	e.g. zinc + acid becomes warm (exothermic)
■ Evolution of a Gas	e.g. H■ gas when zinc reacts with acid
■ Change in State	e.g. solid Mg turns to white powder MgO
■■ Formation of Precipitate	e.g. white BaSO■ from barium chloride + sodium sulphate

2. Chemical Equations

A **chemical equation** is a symbolic representation of a chemical reaction using chemical formulae of reactants and products.

2.1 Word Equation

The simplest way to represent a reaction — written in words:

Word Equation (Activity 1.1)

Magnesium + Oxygen → Magnesium Oxide

2.2 Skeletal Chemical Equation

Uses chemical symbols/formulae instead of words. May be *unbalanced* — not satisfying the law of conservation of mass.

Skeletal (Unbalanced) Equation

$$\mathbf{Mg + O_2 \rightarrow MgO}$$

2.3 Balanced Chemical Equation

A balanced equation has an **equal number of atoms of each element** on both sides. This is required by the **Law of Conservation of Mass**: mass can neither be created nor destroyed in a chemical reaction.

Steps to Balance an Equation (Hit-and-Trial Method):

Step 1	Write the skeletal equation and draw boxes around each formula.
Step 2	Count atoms of each element on LHS and RHS.
Step 3	Start with the compound having the maximum number of atoms. Balance the element with the most atoms first.
Step 4	Balance remaining elements one by one using appropriate coefficients.
Step 5	Check: count all atoms on both sides to verify balance.
Step 6	Add physical state symbols: (s) solid, (l) liquid, (g) gas, (aq) aqueous.
Step 7	Indicate reaction conditions (heat, light, catalyst, pressure) above/below the arrow if needed.

Worked Example — Balancing Fe + H■O → Fe■O■ + H■:

Element	LHS (initial)	RHS (initial)	Action Taken
Fe	1	3	Put coefficient 3 before Fe on LHS
O	1	4	Put coefficient 4 before H■O on LHS
H	2 (in H■O)	2 (in H■)	Put coefficient 4 before H■ on RHS

Final Balanced Equation

$$\mathbf{3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)}$$

■ **Note:** Physical state symbols: (s) = solid, (l) = liquid, (g) = gas, (aq) = dissolved in water. Reaction conditions like temperature, pressure, or catalyst are written above/below the arrow.

3. Types of Chemical Reactions

Reaction Type	Pattern	Key Feature
Combination	A + B → AB	Two/more reactants → one product
Decomposition	AB → A + B	One reactant → two/more products
Displacement	A + BC → AC + B	More reactive element displaces less reactive
Double Displacement	AB + CD → AD + CB	Exchange of ions between two compounds
Redox	Oxidation + Reduction	Simultaneous gain & loss of oxygen/hydrogen

3.1 Combination Reactions

Two or more substances (elements or compounds) combine to form a **single product**. Many combination reactions are also **exothermic** (release heat).

Quick lime + Water → Slaked lime (Activity 1.4)

$$CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) + Heat$$

Burning of coal

$$C(s) + O_2(g) \rightarrow CO_2(g)$$

Formation of water

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$

Burning of natural gas

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$$

Burning of Mg ribbon

$$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$$

■■ **Note:** Exothermic reactions release energy as heat. Slaked lime solution is used for whitewashing. It reacts with CO■ in air to form CaCO■, giving a shiny finish. Marble also has the formula CaCO■.

3.2 Decomposition Reactions

A **single reactant** breaks down into two or more simpler products. These reactions are the **opposite of combination reactions.** Energy must be supplied — as heat, light, or electricity.

Energy Source	Reaction Name	Example
Heat (Δ)	Thermal Decomposition	CaCO■ → CaO + CO■ 2FeSO■ → Fe■O■ + SO■ + SO■ 2Pb(NO■)■ → 2PbO + 4NO■ + O■
Light (hv)	Photo Decomposition	2AgCl → 2Ag + Cl■ 2AgBr → 2Ag + Br■ (used in B&W photography)
Electricity	Electrolytic Decomposition	2H■O → 2H■ + O■ (electrolysis of water)

Thermal decomposition of ferrous sulphate (green → reddish brown)

$$2FeSO_4(s) \xrightarrow{Heat} Fe_2O_3(s) + SO_2(g) + SO_3(g)$$

Photo decomposition of silver chloride (white → grey)

$$2AgCl(s) \xrightarrow{Sunlight} 2Ag(s) + Cl_2(g)$$

■ **Note:** Endothermic reactions absorb energy from surroundings. All decomposition reactions are endothermic because they need an energy input to break bonds.

3.3 Displacement Reactions

A **more reactive element** displaces a **less reactive element** from its compound in solution. Only metals higher in the reactivity series can displace those below them.

Iron displaces copper from CuSO■ (Activity 1.9 — nail turns brownish)

$$Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$$

Zinc displaces copper

$$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$$

Lead displaces copper

$$Pb(s) + CuCl_2(aq) \rightarrow PbCl_2(aq) + Cu(s)$$

3.4 Double Displacement Reactions

Two compounds react by **exchanging their ions**, forming two new compounds. Often one product is an **insoluble precipitate**. These are also called **precipitation reactions**.

White precipitate of barium sulphate is formed (Activity 1.10)

$$Na_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4(s)\downarrow + 2NaCl(aq)$$

Yellow precipitate of lead iodide formed (Activity 1.2)

$$Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s)\downarrow + 2KNO_3(aq)$$

> ■ **Note:** The ↓ symbol indicates a precipitate (insoluble solid that settles). The ↑ symbol indicates a gas that escapes from the solution.

3.5 Oxidation and Reduction (Redox Reactions)

Term	Gains	Loses	Example
Oxidation	Oxygen / Loses Hydrogen	Hydrogen / Gains Oxygen	Cu → CuO (gains O) CuO+H■→Cu+H■O (CuO loses O → Redu
Reduction	Hydrogen / Loses Oxygen	Oxygen / Gains Hydrogen	CuO → Cu (loses O) H■ → H■O (gains O → Oxidation)

Redox Reaction: CuO is REDUCED (loses O); H■ is OXIDISED (gains O)

$$CuO(s) + H_2(g) \xrightarrow{Heat} Cu(s) + H_2O(l)$$

C is oxidised to CO; ZnO is reduced to Zn

$$ZnO + C \rightarrow Zn + CO$$

> ■ **Note:** In a redox reaction, oxidation and reduction ALWAYS occur simultaneously. The substance that is oxidised is the reducing agent; the substance that is reduced is the oxidising agent.

4. Exothermic and Endothermic Reactions

	Exothermic Reactions	Endothermic Reactions
Energy	Released to surroundings	Absorbed from surroundings
Temperature change	Increases (feels hot)	Decreases (feels cold)
Examples	• Burning of fuels • Respiration (C■H■■O■ + 6O■ → 6CO■ + 6H■O + energy) • CaO + H■O → Ca(OH)■ • Decomposition of compost	• Decomposition reactions (thermal, photo, electrolytic) • Ba(OH)■ + NH■Cl (cool to touch) • Electrolysis of water

> ■ **Note:** Respiration is exothermic: Glucose (from food) reacts with oxygen in body cells, releasing energy needed for life processes. C■H■■O■ + 6O■ → 6CO■ + 6H■O + energy

5. Effects of Oxidation in Daily Life

5.1 Corrosion

When metals react with substances in the environment (moisture, oxygen, acids), they are attacked — this is called **corrosion**. It weakens structures and causes enormous economic damage.

Metal	Corrosion Product	Colour of Coating
Iron (Fe)	Hydrated iron(III) oxide — Rust	Reddish brown
Copper (Cu)	Basic copper carbonate	Green
Silver (Ag)	Silver sulphide	Black

■ **Note:** Prevention of corrosion: painting, galvanising (zinc coating), alloying, and electroplating create barriers that prevent moisture and oxygen from reaching the metal surface.

5.2 Rancidity

When fats and oils in food are oxidised (especially by oxygen in air), they develop an unpleasant smell and taste — this is called **rancidity**. It reduces the nutritional value and palatability of food.

Prevention Method	How It Works
Antioxidants (e.g., BHA, BHT)	Added to food to prevent oxidation of fats
Airtight containers / vacuum packing	Reduces exposure to oxygen
Nitrogen flushing in chip packets	Inert N■ replaces O■, preventing oxidation
Refrigeration	Slows down chemical reactions including oxidation

6. Key Definitions at a Glance

Chemical Reaction	A process in which the nature and identity of at least one substance changes, resulting in the formation of new substances.
Chemical Equation	A symbolic representation of a chemical reaction using chemical formulae of reactants and products.
Reactants	Substances that take part in and undergo change during a chemical reaction (written on LHS).
Products	New substances formed as a result of a chemical reaction (written on RHS).
Balanced Equation	An equation in which the number of atoms of each element is equal on both sides.
Law of Conservation of Mass	Mass can neither be created nor destroyed in a chemical reaction.
Combination Reaction	Two or more reactants combine to form a single product.
Decomposition Reaction	A single reactant breaks down into two or more simpler products.
Displacement Reaction	A more reactive element displaces a less reactive element from its compound.

Double Displacement Reaction	Two compounds react by exchanging their ions.
Precipitation Reaction	A reaction that produces an insoluble salt (precipitate).
Exothermic Reaction	A reaction that releases energy (heat) to the surroundings.
Endothermic Reaction	A reaction that absorbs energy from the surroundings.
Oxidation	Gain of oxygen OR loss of hydrogen by a substance.
Reduction	Loss of oxygen OR gain of hydrogen by a substance.
Redox Reaction	A reaction involving simultaneous oxidation and reduction.
Corrosion	Gradual destruction of a metal by its reaction with moisture, oxygen, or other substances in the environment.
Rancidity	The unpleasant change in smell and taste of fats/oils due to oxidation.

7. Important Chemical Equations Summary

Type	Reaction Name	Balanced Equation
Combination	Burning of Mg	$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$
Combination	Quick lime + Water	$CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq)$ + Heat
Combination	Whitewashing reaction	$Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s) + H_2O(l)$
Combination	Respiration	$C_6H_{12}O_6(aq) + 6O_2(aq) \rightarrow 6CO_2(aq) + 6H_2O(l)$ + energy
Decomposition (Heat)	Ferrous sulphate	$2FeSO_4(s) \rightarrow Fe_2O_3(s) + SO_2(g) + SO_3(g)$
Decomposition (Heat)	Calcium carbonate	$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$
Decomposition (Heat)	Lead nitrate	$2Pb(NO_3)_2(s) \rightarrow 2PbO(s) + 4NO_2(g) + O_2(g)$
Decomposition (Light)	Silver chloride	$2AgCl(s) \rightarrow 2Ag(s) + Cl_2(g)$
Decomposition (Light)	Silver bromide	$2AgBr(s) \rightarrow 2Ag(s) + Br_2(g)$
Decomposition (Electricity)	Electrolysis of water	$2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$
Displacement	Fe + CuSO■	$Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$
Displacement	Zn + CuSO■	$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$
Double Displacement	BaCl■ + Na■SO■	$Na_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4(s)\downarrow + 2NaCl(aq)$
Redox	CuO + H■	$CuO(s) + H_2(g) \rightarrow Cu(s) + H_2O(l)$
Redox	Zn + H■SO■	$Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$

8. Quick Revision — Points to Remember

1	A chemical equation must always be balanced (Law of Conservation of Mass).
2	Physical states (s), (l), (g), (aq) make equations more informative.
3	Combination reactions: A + B $\rightarrow$ AB. Often exothermic.
4	Decomposition reactions: AB $\rightarrow$ A + B. Always endothermic (need energy input).
5	In displacement: a more reactive element displaces a less reactive one.
6	Double displacement = exchange of ions; often forms a precipitate.
7	Oxidation = gain of O or loss of H. Reduction = loss of O or gain of H.
8	In redox reactions, oxidation and reduction occur simultaneously.
9	Exothermic: heat released. Endothermic: heat absorbed.
10	Respiration is exothermic. Decomposition reactions are endothermic.
11	Corrosion = slow destruction of metals by oxidation in the environment.
12	Rancidity = oxidation of fats/oils; prevented by antioxidants, N■ flushing, airtight storage.
13	Silver salts (AgCl, AgBr) decompose in light — used in black and white photography.
14	Slaked lime [Ca(OH)■] solution is used for whitewashing; reacts with CO■ to form CaCO■.